

Business Plan: EduPulse AI

Next-Generation Adaptive STEM Copilot

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1 Executive Summary

EduPulse AI is a specialized EdTech startup providing an LLM-based multi-agent system designed to reduce student attrition in higher education STEM fields. Unlike general-purpose AI, EduPulse AI dynamically adjusts the cognitive load of technical content through a proprietary **Adaptive Scaffolding Engine**, filling the gap between massive open online courses (MOOCs) and expensive human tutoring.

2 Product & Technical Core

Our product is a localized, low-latency educational copilot built on a fine-tuned **Qwen2.5-0.5B** architecture. The technical superiority stems from three pillars:

2.1 Multi-Agent Orchestration (The Socratic Pipeline)

Instead of a single-prompt response, EduPulse utilizes a **Coordinator-Worker** architecture:

- **Coordinator Agent:** Performs real-time intent classification and maintains a persistent **Student Persona Vector** (tracking mastery level and sentiment).
- **Tutor Agent:** Specialized in pedagogical transformation. It uses the student's vector to decide whether to employ analogical reasoning (for beginners) or first-principles derivation (for advanced users).
- **Examiner & Reviewer Agents:** An internal adversarial loop where the Examiner generates quizzes and the Reviewer validates them against the target difficulty, ensuring no "Difficulty Drift" occurs.

2.2 Efficient Post-Training (LoRA & DPO)

To ensure the model can be deployed on student-grade hardware (16GB RAM), we employ:

- **SFT (Supervised Fine-Tuning):** Using an **Evol-Instruct** dataset to inject pedagogical depth.
- **DPO (Direct Preference Optimization):** A reference-free alignment strategy that optimizes for "encouraging and guiding" feedback rather than "direct answer-giving," effectively modeling the behavior of elite tutors.

2.3 Technical Implementation

- **Optimization:** Implemented with **Gradient Checkpointing** and **FP16 precision** to minimize memory footprint on Apple MPS and NVIDIA CUDA devices.
- **Feedback Loop:** A RAG (Retrieval-Augmented Generation) layer integrated with university-specific textbooks to ensure factual grounding and eliminate hallucinations.

3 Market Analysis & Competition

We target the **\$8.4 billion** AI-in-Education market, focusing specifically on quantitative disciplines (Math, Physics, CS).

Table 1: Competitive Landscape Analysis

Feature	EduPulse AI	Khanmigo	ChatGPT (OpenAI)
Technical Logic	Multi-Agent Adaptive Loop	Single-Model Scripting	General Purpose
Difficulty Scaling	Dynamic (Real-time)	Linear (Fixed path)	Inconsistent
Deployment	Edge-compatible (LoRA)	Cloud-only	Cloud-only
STEM Focus	High (Symbolic reasoning)	Medium (K-12 focus)	Low (General)

4 Problem & Evidence

- **The Problem:** 40% of STEM students drop out due to the “Pacing Gap”—where lecture speed exceeds their individual processing rate.
- **The Evidence:** Our internal research (Q1 Synthesis) demonstrated that beginner-level students require **3.8x more intermediate steps** to solve the same Calculus problem compared to advanced students. EduPulse AI identifies this gap and “stretches” the explanation accordingly.

5 Management & Objectives

- **Team:** Comprised of LLM Engineers specializing in PEFT (Parameter-Efficient Fine-Tuning) and Educational Psychologists.
- **Milestones:**
 - **Phase 1:** Beta testing in 3 major University CS departments.
 - **Phase 2:** Scaling the Multi-Agent framework to support 50+ STEM sub-topics.
 - **Phase 3:** Integration with Learning Management Systems (Canvas/Blackboard).

6 Risk Mitigation

- **Hallucination Risk:** Managed by the **Reviewer Agent** which cross-references outputs against verified academic knowledge bases.
- **Hardware Constraints:** Ongoing optimization of 0.5B - 1.5B parameter models to ensure smooth performance on standard student laptops.